

AI/ML Task 2: Feature Engineering, Model Optimization & Performance Comparison

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Dataset: California Housing dataset from scikit-learn

Evaluation protocol: 80% training data and 20% held-out test data, `random_state=42`

Objective and approach

This project extends the Task 1 house-price predictor by adding feature engineering, leakage-safe feature scaling, and comparison of multiple regression algorithms. The target variable, `HousePrice`, represents median house value in units of \$100,000. The dataset contains 20,640 rows, eight original numeric predictors, and no missing values.

Two engineered features were added: `RoomsPerPerson`, which relates average rooms to average occupancy, and `BedroomsPerRoom`, which describes the proportion of bedrooms among average rooms. These ratios expose relationships that are not represented directly by the original columns. The final feature matrix therefore contains 10 predictors.

The data was divided into training and test sets using an 80:20 split. `StandardScaler` was placed inside each modeling pipeline and fitted only on the training data. This prevents information from the held-out test set from influencing preprocessing. Three models were trained using the same split: Linear Regression, Ridge Regression with `alpha=1.0`, and a Decision Tree Regressor with `max_depth=5`.

Model comparison

Model	MAE	RMSE	R ²
Linear Regression	0.4862	0.6753	0.6519
Ridge Regression	0.4862	0.6754	0.6519
Decision Tree	0.5214	0.7240	0.6000

Lower MAE and RMSE indicate smaller prediction errors, while higher R² indicates that more target variation is explained. Linear Regression is the best model by RMSE and R², although Ridge Regression is almost identical. This result suggests that the engineered features improve the linear baseline and that the selected regularization strength does not materially change

performance. The depth-limited Decision Tree performs worse on this split, indicating that a shallow tree does not capture the continuous relationships as effectively as the linear models.

Best-model validation

The selected model is Linear Regression. Its MAE of 0.4862 corresponds to an average absolute error of approximately USD 48,616 because the target is measured in \$100,000 units. Its RMSE is 0.6753, and it explains approximately 65.2% of the variance in the held-out target values. The actual-versus-predicted and residual plots are included in the notebook for visual validation. They confirm that the model captures the broad trend but remains a baseline affected by the target cap and geographic effects.

Conclusion and improvement ideas

Feature engineering and leakage-safe scaling improved the Task 1 workflow. Among the tested models, Linear Regression is the best choice for this experiment because it produced the lowest RMSE and highest R^2 while remaining highly interpretable. Ridge provides nearly identical performance and may be preferable when coefficient stability or stronger regularization is required.

Further optimization should use cross-validation and a systematic hyperparameter search. Useful experiments include tuning the Ridge penalty, testing deeper or pruned trees, evaluating Random Forest and Gradient Boosting models, and adding nonlinear geographic features. Performance should also be measured separately across regions and price ranges because overall metrics can hide uneven behavior.

The selected model was saved as `best_house_price_model.joblib`. The executed notebook contains the complete implementation, model comparison table, charts, and final selection logic.

References